HORIZON EUROPE FRONTIER RESEARCH

Mastering the ERC Grant Application

Practical guide to proposal design, budget calculation, portal submission, and host institution commitment (High-Energy Physics / PE2 edition)

00 / GRANT SCHEMES OVERVIEW (2026–2027)

Core ERC Individual Funding Schemes

Plus +€1.0M standard additional funding (or +€2.0M when relocating from a third country) for StG, CoG, and AdG.

€1.5M

Starting Grant (StG)

0–10 years post-PhD duration up to 5 years

€2.0M

Consolidator Grant (CoG)

5–15 years post-PhD duration up to 5 years

€2.5M

Advanced Grant (AdG)

10–year track record duration up to 5 years

€7.0M

ERC Plus Grant (New)

Up to 7 years (~30 grants/year across all stages)

SECTION 01

01 Proposal Preparation

01 / PROPOSAL PREPARATION

Three-Part Architecture & Two-Step Evaluation

Critical Rule: Step 1 panel generalists read ONLY Part B1; Part B2 is opened solely in Step 2.

A

Part A: Online Forms

Administrative data panel descriptors budget table ethics

B1

Part B1: Step 1 File

5–page extended synopsis plus 4–page PI CV

B2

Part B2: Step 2 File

7–page full scientific proposal methodology resources

HI

Annexes

Host Institution support letter PhD certificate extensions

Tailoring Part B1 vs. Part B2 to PE2 Evaluators

Part B1: Extended Synopsis (5 pp.)

Target Audience: Broad PE2 panel (nuclear, string, neutrino, astro, lattice).

Core Focus: Fundamental physics question and paradigm shift.

Writing Style: Avoid uncontextualized collider or loop-integral shorthand.

Feasibility Pitch: Concise proof of concept without micro-details.

Part B2: Scientific Proposal (7 pp.)

Target Audience: Specialist HEP remote referees plus Step 2 panel.

Core Focus: Deep methodology work packages and risk mitigation.

Writing Style: Rigorous technical execution and quantitative fallbacks.

Group Boundary: Isolate PI group tasks from collaboration service work.

Part B1 Synopsis: Tips, Tricks & Fatal Pitfalls

Step 1 Winning Tips

Page 1 Hook: Frame fundamental physics bottleneck on page one.

Concept Figure: Include one clear visual architecture diagram.

Self-Contained Pitch: Summarize method feasibility inside the 5 pages.

Non-Expert Test: Have a colleague outside your subfield review it. Avoid jargon!

Common B1 Pitfalls to Avoid

Page 1 Acronym Soup: Opening with raw N3LO or Phase-2 trigger jargon.

Cross-Referencing B2: Never write "see Part B2" for core methods.

3-Page Background: Avoid textbook SM reviews before stating your aims.

Unbacked Claims: Never promise paradigm shifts without concrete steps.

"PE2 Panel" Reality Check: NOT just collider phenomenologists or LHC experimentalists. It includes nuclear physicists, string theorists, lattice QCD specialists, neutrino physicists, and gravitational-wave / astroparticle researchers.

Part B2 Scientific Proposal: HEP Architecture & Traps

Tip: Non-Domino Work Packages

Ensure WP1 failure never blocks WP2 or WP3

Tip: Quantitative HEP Decision Gates

Trigger numeric/asymptotic or sensor Plan B at Month 18

Pitfall: Crank-Turning & Blind Scans

Avoid routine loop/leg additions or luminosity-only limit updates

Pitfall: Collaboration Service Work

Separate PI frontier innovation from routine ATLAS/CMS MoU tasks

Crafting the Scientific Case & PI Narrative

2,000-Char Abstract: State physics bottleneck methodology and breakthrough impact immediately.

High-Risk / High-Gain: Guarantee major precision or tool impact even without a BSM signal.

4-Page CV & Track Record: Contextualize 10 key papers or community software packages.

Intellectual Independence: Prove clear separation from former PhD/postdoc mentors or giant consortia.

Why Now & Why You: Connect new theoretical/detector methods to your unique leadership.

01 / SCIENTIFIC CASE DEEP DIVE

Calibrating High-Risk / High-Gain in Post-Higgs HEP

Toy-Limit / Test-Beam Pilot

De-risk via planar/massless limit or prototype test-beam data.

The Crank-Turning Trap

Prove why old formalisms break and require new mathematics.

Signal-Independent Gain

Deliver lasting SMEFT bounds or public MC tools without BSM discovery.

4-Page Narrative CV: HEP Best Practices & Red Flags

CV Tips & Tricks (CoARA Rules)

Unpack 3,000-Author Papers: Cite analysis contact, internal notes, and L2/L3 convener roles.

Software as First-Class Output: Highlight MC generators, amplitude libraries, and adoption.

Show Theory Independence: Feature papers with junior mentees rather than former advisors.

Explain Career Breaks: Clearly document parental, clinical, or medical leaves.

Track Record Red Flags

Journal Impact Factors: Strictly banned by ERC; cite article-level or community uptake.

Opaque Alphabetical Lists: Listing collaboration papers without stating your exact role.

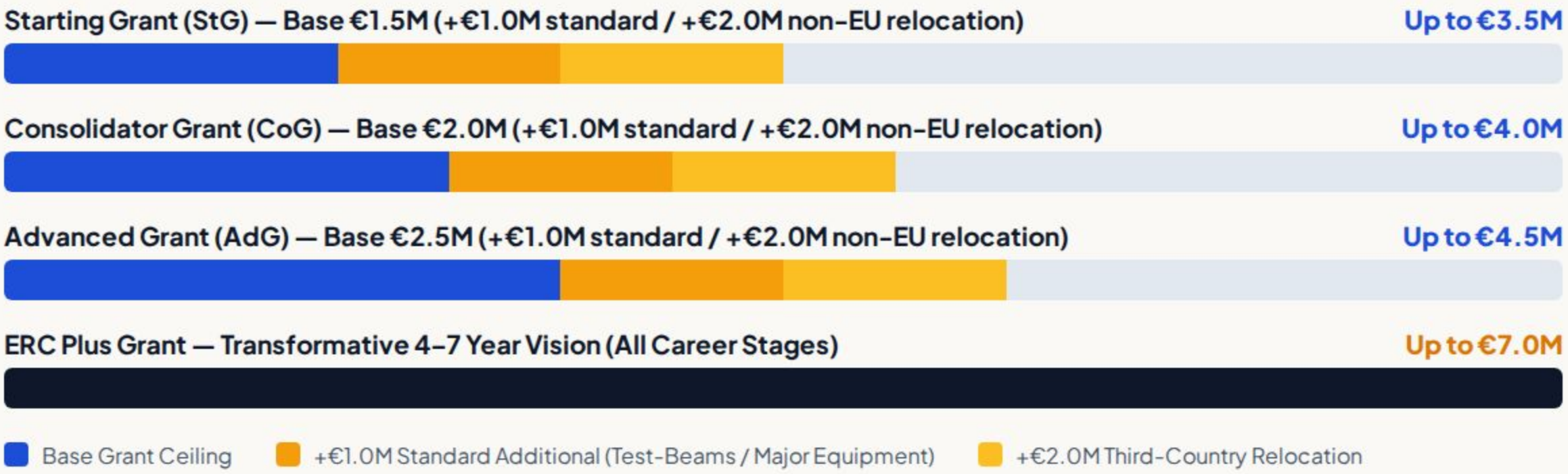
Former Advisor Co-Authorship: Exclusively publishing small theory papers with mentors.

Skill-Project Mismatch: Proposing ML or hardware tasks absent from your track record.

SECTION 02

02 Budget

Maximum Grant Ceilings & Additional Funding



02 / BUDGET STRUCTURE (HEP SPECIFICS)

Cost Categories, HPC/CERN Rules & 25% Overhead

Cost Category	HEP Eligible Items	Overhead Rule	PE2 Justification Focus
A. Personnel	PI salary, postdocs, PhDs, detector engineers	**+25% Indirect**	Match minimum PI time commitment
C1. Travel & LTA	CERN shifts, LTA stays, collaboration weeks	**+25% Indirect**	Justify elevated CERN/test-beam travel
C2. Equipment & HPC	High-RAM/GPU nodes, ASICs, cleanroom rigs	**+25% Indirect**	Depreciation vs custom prototype; why not Grid/EuroHPC
C3. Other Goods	Silicon wafers, cryogens, open-access fees	**+25% Indirect**	Note SCOAP3 covers standard HEP journals
B. Subcontracting	External ASIC foundry or irradiation service	**0% Indirect**	No core scientific tasks outsourced
E. Indirect Costs	**Flat-rate institutional overhead**	**Automatic 25%**	**Included inside grant ceiling**

SECTION 03

03 Funding & Tender System

Five-Step Submission Workflow on the EU Portal

01

EU Login & Call Selection

Create proposal draft under the specific ERC call identifier.

02

Link Host PIC & Officers

Enter 9-digit Host Institution PIC and grant Main Contact access.

03

Download Official Templates

Retrieve call-specific RTF/DOCX templates directly inside the portal workspace.

04

Complete Part A & Validate

Fill online budget and ethics tables then resolve red blocking errors.

05

Submit Early & Overwrite

Click Submit 48h early; re-upload updated PDFs until 17:00:00 Brussels time.

EU Login & Call Selection

Step 01

Initialize Draft Workspace

Verify exact Horizon Europe ERC work programme call identifier before creating your draft.

EU Login Account: Register with institutional email and enable two-factor authentication.

Locate Exact Call ID: Filter Funding & Tenders Portal by ERC-StG, CoG, AdG, or Plus.

Start Submission: Select ERC action type to launch your draft workspace.

Set Acronym & Panel: Choose primary panel (PE2) and cross-panel descriptors (PE9/PE1/PE3).

03 / PORTAL STEP 02

Link Host PIC & Assign Portal Roles

9-Digit Host PIC

Enter validated institutional Participant Identification Code.

Principal Investigator

Link ORCID identifier to PI profile.

Main Host Contact

Grant full read-write access to Grants Office.

03 / PORTAL STEP 03

Download Templates & Enforce Formatting Rules

Mandatory Portal Package

Part B1 Template: Extended synopsis plus 4–page CV and track record.

Part B2 Template: Full 7–page scientific methodology (LaTeX / Word).

Host Support Letter: Official binding statement for institutional signature.

Eligibility Annexes: PhD certificate and career break proof templates.

Strict Typography & PDF Rules

Font Floor: Minimum 11pt Times New Roman or Arial (check LaTeX scaling).

Page Margins: Minimum 2.0 cm side and 1.5 cm top/bottom.

Line Spacing: At least single spacing across all body paragraphs.

Unencrypted PDF: Flatten forms and remove password security locks.

03 / PORTAL STEP 04

Complete Part A Web Forms & Run Validation

General Data & Abstract

2,000-character abstract plus ERC review keywords

Section 3 Budget & Narrative

Direct cost inputs plus 4,000-character justification

Section 4 Ethics & Security

Flag non-EU labs (CERN/Fermilab/KEK) and dual-use items

Validation Check Engine

Clear red blocking errors; review yellow warnings

03 / PORTAL STEP 05

Submit Early & Overwrite Until Deadline

48h

Initial Submission Buffer

Lock in a valid submission before 17:00:00 Brussels cutoff.

Save Does Not Submit: Clicking Save & Close leaves your proposal as an unsubmitted draft.

Unlimited Overwrites: Upload revised PDFs and click Submit again to replace prior versions.

e-Receipt Verification: Download the digitally signed timestamped proof after each submission.

Zero Grace Period: Portal locks automatically at 17:00:00 CET/CEST sharp.

Critical Portal Mechanics & HEP Ethics Checks

PE2 vs. PE9 / PE1 / PE3

Use secondary descriptors for astro/cosmo or math amplitudes.

Strict Page Limits

Portal auto-truncates excess uploaded PDF pages.

HEP Ethics & Dual-Use

Declare non-EU labs (CERN/FNAL) and rad-hard/isotope items.

SECTION 04

04 Host Institution

Host Institution Commitment & In-Kind Support

Mandatory Template: Use the official unaltered ERC Host Institution Binding Statement.

Legal Signatory: Signed by the legal representative (Director, Rector, VP Research, or LEAR).

Scientific Independence: Guarantees PI autonomy to direct research and select team members.

HEP In-Kind Negotiables: Secure local cluster quotas, cleanroom/workshop hours, and buyouts.

Lead Time: Initiate internal institute approval 4–6 weeks before call deadline.



04 / HOST INSTITUTION

Leveraging Additional Host Support Services

Grant Office Pre-Review

Eligibility verification internal peer feedback

Financial & HR Costing

Exact salary scales social security & CERN LTA rates

HEP Open Science & DMP

arXiv/SCOAP3, HEPData, Rivet & Zenodo preservation

Step 2 Mock Interviews

Pitch deck coaching Q&A simulation panels

Useful 'local' contacts:

IFJ PAN Dept. of Foreign, Structural & Domestic Projects Financial Management (DPN): Malgorzata.Borek@ifj.edu.pl & Alicja.Borys@ifj.edu.pl
Preparation support by HPK (pre-review, mock interview): https://www.gov.pl/web/hpkppw & https://transfer.edu.pl/pl/
PAN Excellence in Science Bureau: doskonalosc@pan.pl & https://pan.pl/granty-erc/

SECTION 05

05 Evaluation Process

05 / EVALUATION PROCESS

Evaluation Workflow: From Submission to Grant

Key milestones and decision points across all three panel stages

STAGE 0

Evaluation Preparation

Initial Panel Chairs Meeting: Align scoring benchmarks across panels.

Proposal Allocation: Match proposals to PE2/cross-panel reviewers via descriptors.

STEP 1

Remote Review & Downselection

Remote Part B1 Review: Panel generalists evaluate 5-page synopsis and CV.

Step 1 Panel Meetings: Proposals scored A advance (max 44 per panel).

STEP 2

Full Evaluation & PI Interviews

Full Assessment: Specialist referees + panel review B1, B2, and budget.

PI Interviews: Live Q&A decides final ranking (Funded A, Reserve, B).

05 / EVALUATION PROCESS

What Happens After Submission?

Key evaluation outcomes, penalty thresholds, and resubmission restrictions across ERC calls

Step 2 & High-Pass (No Penalty)

Category A / B at Step 2: Eligible to apply immediately in the next Work Programme.

Category A at Step 1 (not invited): Eligible to resubmit immediately without waiting period.

Step 1 Lockouts (B & C Scores)

Category B at Step 1: 1-year lockout from StG, CoG, AdG, or Plus calls.

Category C at Step 1: 2-year lockout across the next two Work Programmes.

Integrity & Call-Specific Rules

Research Integrity Violations: Minimum 2-year complete ban across all ERC actions.

Synergy & Plus Grants: Separate standalone restriction rules apply per Work Programme.

Questions & Discussion

Start early, engage your Host Institution Grants Office 8 weeks before deadline, and submit a complete draft 48 hours ahead.

More information: https://erc.europa.eu/apply-grant/

Credits: Google Gemini (helping to bring presentation in shape)